

Polynomial Regression Assignment — Report

Roll No. **BT2024154** | Problems: var1 (steam turbine Net Power Score), var2 (thermal anomaly mapping)

Code repository: <https://github.com/Kkshiteej-Tiwari/Machine-Learning-Assignment-1>

1. Summary

Both problems were solved with **polynomial regression only**: every model is a polynomial in the inputs containing all terms of total degree $\leq d$, fitted by penalised least squares. The degree and the penalty were chosen by repeated 5-fold cross-validation (CV). The final choices are **degree 5** for var1 (CV MSE 0.351, CV R^2 0.967) and **degree 8** for var2 (CV MSE 0.247, CV R^2 0.9946). The largest allowed degrees (10 and 20) overfit badly. For var2 the CV error flattens at about 0.25 for degrees 8–10, which looks like the noise floor, so degree 8 is the smallest degree that reaches it.

2. Data exploration

	inputs	train / test rows	Var(y) train	max degree	rows with ≥ 3 inputs at ± 1 (train $\rightarrow$ test)
var1	x1..x6	1000 / 1000	10.49	10	28.7% $\rightarrow$ 66.2%
var2	x1..x3	1000 / 1000	45.52	20	1.5% $\rightarrow$ 1.7%

The files contain no missing values. All inputs lie in $[-1, 1]$, and a large share of values sit *exactly* on ± 1 (31% of var1 training values, 25% for var2). The inputs were evidently clipped to the box during preprocessing. This matters in two ways. (i) High-degree polynomials are least stable near the edges of the domain (Runge-type oscillation), and many points sit on those edges. (ii) For **var1 the test set is shifted toward the corners**: 66% of test rows have three or more clipped inputs, against 29% of training rows. CV on the training rows alone would therefore under-represent the hardest test region. Section 3.3 explains how this is handled. var2 shows no such shift.

3. Approach

3.1 Model family and feature basis

For p inputs and degree d the model is $f(x) = \sum c_t \cdot \phi_t(x)$, summed over all exponent vectors t with $|t| \leq d$, i.e. $C(p+d, d)$ terms. That is 462 terms for var1 at $d=5$ (8008 at $d=10$) and 165 for var2 at $d=8$ (1771 at $d=20$). This matches the assignment's definition: the powers in each term add up to at most d . Rather than raw monomials $x_1^a x_2^b \dots$, I use the **product-Legendre basis** $P_a(x_1)P_b(x_2)\dots$. Every Legendre product expands into monomials of the same total degree, so it spans *exactly the same set of polynomials* and the model is still plain polynomial regression. The design matrix, however, is far better conditioned on $[-1, 1]$. For var2 the condition number at $d=8$ drops from 1.2×10^3 (monomials) to 24, and at $d=12$ from 7.3×10^4 to 290. This keeps high-degree fits numerically reliable, and it makes a penalty on the coefficients mean the same thing for every term.

3.2 Regularisation

A degree- d polynomial with hundreds or thousands of terms fitted to 800 rows (one CV fold) overfits without a penalty. With $d=6$ for var1 there are more terms (924) than rows, so the training error drops to 0 while the CV error explodes (Figure 1). Three penalised estimators were used:

- **Ridge (L2)**, $\min \|y - \Phi c\|^2 + \alpha \|c\|^2$, with an unpenalised intercept. It is solved through one SVD of the centred design matrix per fold, which gives predictions for all 37 α values (10^{-8} – 10^4) at once. Sweeping every degree (10 for var1, 20 for var2) therefore takes about a minute.
- **Lasso (L1) with a degree-weighted penalty**, $\min \frac{1}{2n} \|y - \Phi c\|^2 + \lambda \sum_t (1+|t|)^{\gamma/2} |c_t|$. L1 can set coefficients to exactly zero, which helps when the true polynomial uses only a subset of the possible terms. The weight $(1+|t|)^{\gamma/2}$ penalises high-order terms more than low-order ones, so the fit prefers smooth explanations. It is implemented by rescaling the columns of Φ , and $\gamma=0$ gives ordinary Lasso.
- **Elastic net**, the same objective with the penalty $\lambda[\rho \|c\|_1 + \frac{1}{2}(1-\rho)\|c\|^2]$ (same γ weighting). It sits between ridge ($\rho=0$) and Lasso ($\rho=1$) and is the usual remedy when correlated terms make the Lasso pick among them erratically. $\rho \in \{0.5, 0.9\}$ was tried; small ρ behaves like ridge, which stage 1 already covers.

3.3 Model selection

All hyper-parameters (degree d , penalty α or λ , weight γ) were chosen by 5-fold CV, repeated with two different random splits and averaged (seeded, fully reproducible). Stage 1 runs a ridge sweep over *every* degree to map the bias–variance curve. Stage 2 runs a Lasso grid over the most promising degrees and $\gamma \in \{0, 2, 4, 6\}$. Stage 3 runs elastic net at the degrees and γ values around the Lasso optimum. Because a richer model family can win by chance, elastic net replaces the ridge/Lasso winner only if its per-fold advantage (paired over the same 10 folds) exceeds one standard error. To account for the var1 shift, the selection score is an **importance-weighted CV MSE**. Each validation row with k clipped inputs gets weight $w(k) = P_{\text{test}}(k) / P_{\text{train}}(k)$, so the CV error estimates the error on the test set's mix of rows. Unweighted CV MSE is reported alongside it. For var2 the weights are about 1, and both scores pick the same model.

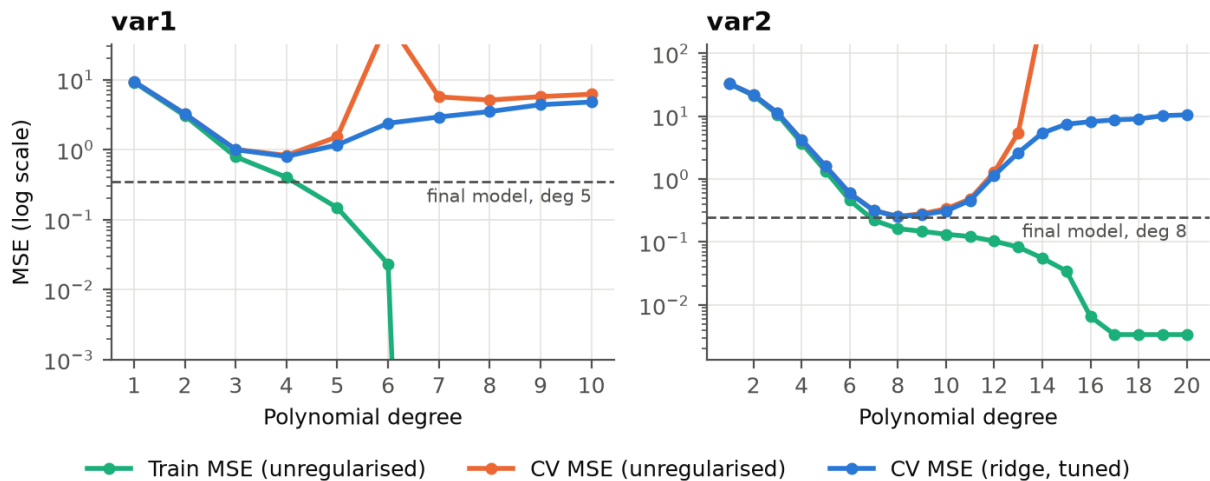


Figure 1. Error against polynomial degree (log scale). Without a penalty the training error keeps falling while the CV error rises sharply past degree 4 (var1) and 9 (var2): the classic overfitting U-curve. Ridge tames the explosion but cannot beat the low-degree optimum. The dashed line is the CV MSE of the final model.

4. Problem var1 — steam turbine Net Power Score (6 inputs, degree ≤ 10)

model	degree	terms (non-zero)	CV MSE	CV R ²	shift-weighted CV MSE
ridge ($\alpha=1e-08$)	2	28	3.236	0.691	5.686
ridge ($\alpha=1e-08$)	3	84	0.999	0.905	1.218
ridge ($\alpha=1e+01$)	4	210	0.800	0.924	0.975
ridge ($\alpha=1e+01$)	5	462	1.166	0.889	1.916
ridge ($\alpha=5e+03$)	6	924	2.383	0.773	2.906
ridge ($\alpha=1e+04$)	10	8008	4.785	0.544	6.710
lasso $\gamma=6$	4		0.619	0.941	0.777
lasso $\gamma=0$	5		0.391	0.963	0.433
lasso $\gamma=6$	6		0.357	0.966	0.388
lasso $\gamma=6$ (final)	5	462 (156)	0.351	0.967	0.381

Chosen degree: 5. Degrees 1–3 clearly underfit: CV MSE falls from 9.34 to 1.00, and train and CV errors stay close together. With ridge the best degree is 4 (CV MSE 0.800), and degree 5 is already worse (1.166). This is because ridge shrinks all 462 coefficients and never removes any, so the extra variance outweighs the gain. The Lasso shows the real structure. The function has genuine degree-5 content but uses only part of the 462 terms: the final model keeps 156 of them, mostly terms that involve one to three inputs. Lasso at degree 5 **more than halves the error** of the best ridge model (0.351 vs 0.800; R² 0.924 $\rightarrow$ 0.967). Going to degree 6 brings no further gain (0.357), and degrees 7–10 only add variance. The degree-weighted penalty ($\gamma>0$) improves on plain Lasso (0.391 $\rightarrow$ 0.351) because it lets the high-order terms in only when the data supports them, which keeps the surface smooth at the clipped corners the test set is rich in. Degree 5 is also best on the unweighted CV score (lowest 0.350), so the choice of degree does not depend on the shift correction.

Robustness at the corners. The out-of-fold MSE of the final model grows only mildly with the number of clipped inputs (0.27 with none, 0.36 with three, 0.41 with five). The expected test MSE is therefore close to the weighted

CV value of about 0.38, not far above it. The final model's test predictions differ from those of the runner-up (degree 6 Lasso) by a mean squared difference of only 0.002, so they are stable.

5. Problem var2 — thermal anomaly mapping (3 inputs, degree ≤ 20)

model	degree	terms (non-zero)	CV MSE	CV R ²	train MSE (no penalty)
ridge ($\alpha=1e+01$)	4	35	4.202	0.9077	3.732
ridge ($\alpha=1e+01$)	6	84	0.594	0.9870	0.460
ridge ($\alpha=5e+00$)	7	120	0.316	0.9931	0.222
ridge ($\alpha=5e+00$)	8	165	0.253	0.9944	0.163
ridge ($\alpha=1e+01$)	9	220	0.271	0.9940	0.148
ridge ($\alpha=5e+00$)	10	286	0.311	0.9932	0.132
ridge ($\alpha=2e+00$)	12	455	1.144	0.9749	0.103
ridge ($\alpha=2e+01$)	14	680	5.395	0.8815	0.055
ridge ($\alpha=2e+03$)	20	1771	10.500	0.7693	0.003
elastic net $\gamma=4$ (final)	8	165 (164)	0.247	0.9946	0.166 (penalised)

Chosen degree: 8. The CV curve (Figure 1, right) is a clean U. Error drops steeply up to degree 7–8, stays flat at about 0.25 for degrees 8–10, then rises fast: about 1.1 at degree 12, 5.4 at degree 14 and over 10 at degree 20. The flat bottom suggests the remaining error is mostly measurement noise. The training error at degree 8 (0.163) is only modestly below the CV error (0.253), which fits a noise variance of roughly 0.2. A higher degree cannot reduce that noise; it can only fit it. By the usual parsimony rule (the smallest model whose CV error matches the minimum), degree 8 is chosen. The “up to degree 20” in the brief is an upper bound on the true complexity, not a target. With 1000 samples in 3-D, a degree-20 model has 1771 coefficients, more than the number of points, and even heavy ridge regularisation leaves its CV error 40 times worse.

At degree 8 ridge, Lasso and elastic net are practically tied (CV MSE 0.253, 0.249 and 0.247). The test predictions of the final model and the ridge model differ by a mean squared difference of only 0.004, so the choice between them is immaterial. The final model (elastic net, $\gamma=4$, $\rho=0.5$) was kept because it had the lowest CV score and passed the one-standard-error test of Section 6. It is nearly dense: only 1 of the 165 terms are zero, which fits a smooth field that genuinely uses almost every degree-8 term.

6. Elastic net check

	model	degree	γ	ρ (L1 share)	λ	shift-weighted CV MSE
var1	best ridge	4	—	0	1.0e+01	0.9747
	best Lasso	5	6	1	2.2e-04	0.3814
	best elastic net	5	4	0.5	1.0e-03	0.4921
	best elastic net	5	4	0.9	1.0e-03	0.3989
elastic net gain over Lasso						-0.0174 $\pm$ 0.0079 $\rightarrow$ not adopted
var2	best ridge	8	—	0	4.6e+00	0.2520
	best Lasso	8	2	1	1.0e-03	0.2478
	best elastic net	8	4	0.5	6.8e-05	0.2459
	best elastic net	9	4	0.9	1.5e-04	0.2463
elastic net gain over Lasso						+0.0020 $\pm$ 0.0014 $\rightarrow$ adopted

Elastic net mixes the two penalties, so it was tested to make sure the choice between ridge and Lasso was not leaving accuracy on the table. The last row of each block is the mean per-fold improvement of the best elastic net over the selected ridge/Lasso model, $\pm$ one standard error. In short, for var1 elastic net does not beat the Lasso model (-0.0174 ± 0.0079), so it was not adopted; for var2 elastic net improves the shift-weighted CV MSE by 0.0020, more than one standard error (0.0014), so it was adopted. The pattern is plausible. var1's function is sparse (the Lasso zeroes two thirds of the terms), so the L2 part only pulls in terms that should be absent. var2's field uses almost every term, so a partly-L2 penalty that shrinks without zeroing suits it slightly better. The var2 gain is small (under 1% of the error) and changes the test predictions by a mean squared difference of only 0.002; the degree choice is unaffected either way.

7. Final models and validation

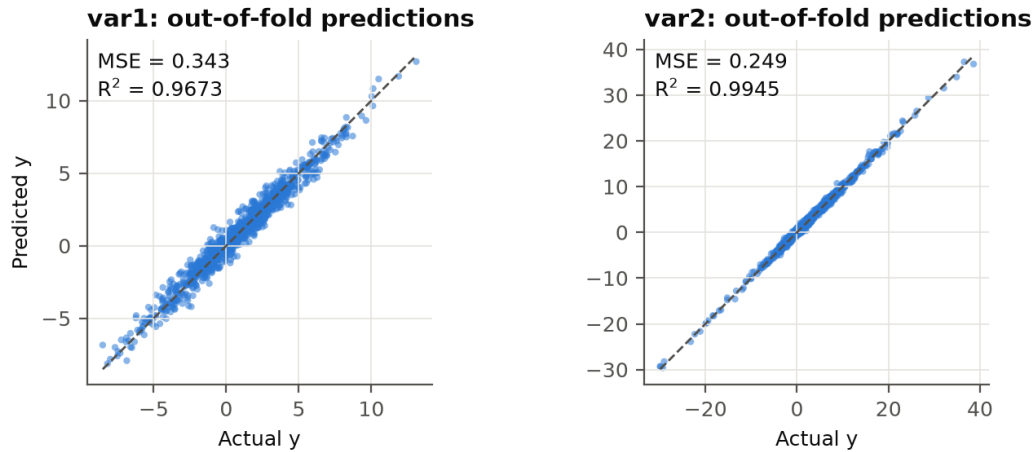


Figure 2. Out-of-fold predictions of the final models against the actual values. No row is predicted by a model that saw it during training.

	degree	method	penalty	non-zero terms	CV MSE	CV R ²	train MSE
var1	5	Lasso, $\gamma=6$	$\lambda=2.2\text{e-}04$	156 / 462	0.351	0.9665	0.238
var2	8	elastic net, $\gamma=4$, $\rho=0.5$	$\lambda=6.8\text{e-}05$	164 / 165	0.247	0.9946	0.166

Each final model was refitted on all 1000 training rows with the selected hyper-parameters and used to predict the test rows. The predictions are in **BT2024154_pred_var1.csv** and **BT2024154_pred_var2.csv**, each a single column y with 1000 rows in the same order as the test files, matching the sample submission.

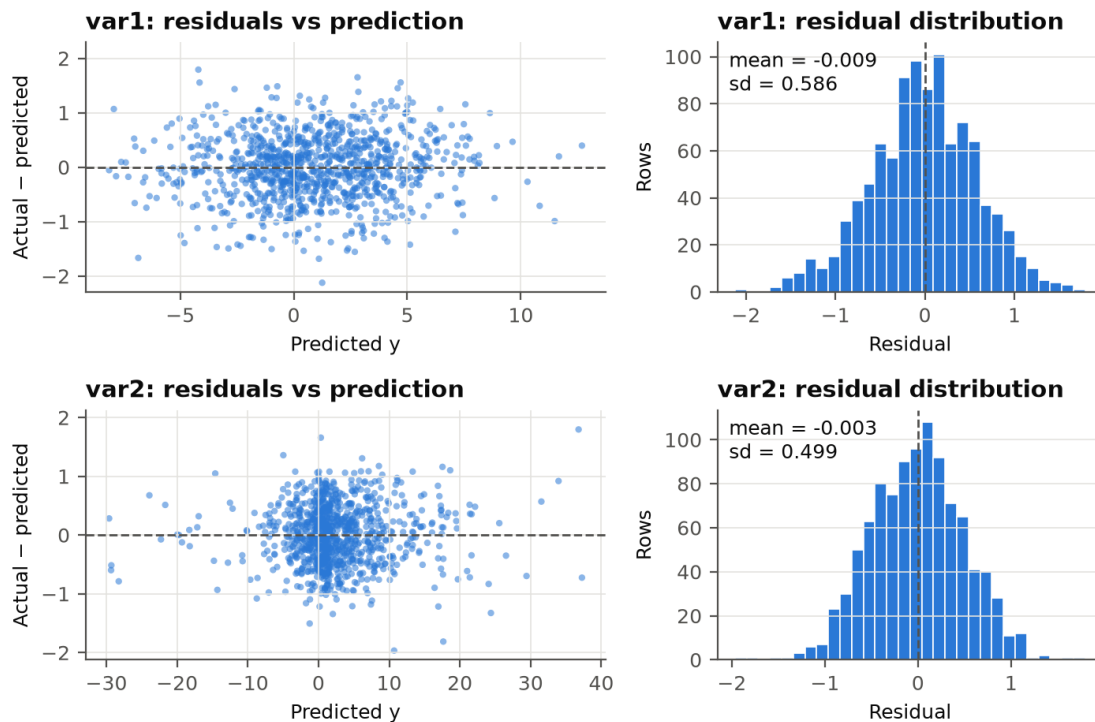


Figure 3. Out-of-fold residuals (actual – predicted) of the final models: against the predicted value (left) and their distribution (right).

Residual checks. For both problems the residuals are centred on zero (mean -0.009 for var1, -0.003 for var2), so the models are not biased. They show no curve or trend against the prediction, so no systematic structure is left for a higher degree to capture. Their spread is the same across the whole range of y, with no funnel shape, so the error is not concentrated in large or small outputs. The distributions are roughly bell-shaped with standard deviations of 0.59 (var1) and 0.50 (var2), consistent with the noise levels estimated in Sections 4 and 5. With Figure 1, this shows the models fit the signal without overfitting.

8. Key decisions at a glance

- **CV over training error** for choosing the degree: training error always favours the highest degree (Figure 1).
- **Legendre basis**: the same polynomial space as monomials, but well conditioned, so degree 8–20 fits are numerically trustworthy.
- **Regularisation**: needed once the number of terms approaches the number of samples. The sparse, degree-weighted Lasso was the decisive improvement for var1. Elastic net was tested as well and adopted only where it beat the winner by more than one standard error (var2, by a small margin).
- **Shift-aware validation**: var1's test set is concentrated at clipped corners, so CV errors were re-weighted to match it, and the chosen model was checked to hold up there.
- **Parsimony**: where several degrees tie within noise (var2: 8–10), the smallest was chosen.

9. Reproducing the results

Repository: <https://github.com/Kkshiteej-Tiwari/Machine-Learning-Assignment-1>. Run `pip install -r requirements.txt`, then `python code/train.py` (model selection and fitting, about 5 minutes on a laptop CPU), then `python code/predict.py` (writes the prediction files). `code/polyreg.py` holds the feature construction, the ridge path, Lasso and elastic-net CV and the final model class. Full CV logs are in `results/`.